Free Majorana Model

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1 Model at hand

1.1 Hamiltonian

$$H = \tfrac{1}{N} \sum_{k=1}^{N} \frac{1 - s_k i d_{k0} d_{k1}}{2} \tag{1}$$

This Hamiltonian is a sum of Hermitian squares (SOHS)[1], its coefficients sum to 1. The operators $id_{k0}d_{k1}$ have eigenvalues ± 1, while s_k are fixed signs.

2 Goal and Method

For eq. (1), the gap is given by $\frac{1}{N}$. We now want to maximize this gap by adding permutation-invariant higher-body interaction terms to the Hamiltonian. In order to do so, we first introduce some new notations:

- $X_k = s_k i d_{k0} d_{k1}$.
- S is a (sub)set of the N modes: $S \subseteq \{1, 2, ..., N\}$. Its cardinality is denoted by $|S|$.
- $X_S = \prod_{k \in S} X_k$ is a $|S|$-body operator.

The generic form of the new Hamiltonian is now

$$H(\vec{c}) = \sum_S c_S \frac{1 - X_S}{2}. \tag{2}$$

By our normalization convention, we now must have $\sum_S c_S = 1$ and $c_S \geq 0$. The sum runs over all nonempty subsets $S \subseteq \{1, \ldots, N\}$ satisfying $|S| \leq k_{\max}$.

[1] Squaring the terms explicitly leaves the terms invariant.

3 Expression for the Energy

We assume our model to be invariant under permutations of modes. This allows us to assume all c_S with equal $|S|$ are equal. Assume we have a set S with $|S| = k$. There are $\binom{N}{k}$ such sets. Introducing

$$c_k = \sum_{|S|=k} c_S = \binom{N}{k} c_S. \tag{3}$$

The normalization condition now reads

$$\sum_{k=1}^{k_{\max}} c_k = 1. \tag{4}$$

The Hamiltonian can now be written as

$$H(\vec{c}) = \sum_{k=1}^{k_{\max}} \frac{c_k}{\binom{N}{k}} \sum_{|S|=k} \frac{1 - X_S}{2}. \tag{5}$$

To find the energy, we act with the Hamiltonian on a state. All operators X_k commute, so the Hamiltonian is diagonal in their common eigenbasis. Due to permutation invariance, all eigenstates containing the same number n of excited modes have the same energy. The energy can therefore be labeled by n. For given coefficients c_k, a maximal interaction order $k_{\max}$, the energy of a state with n excitations is given by

$$E(n, k_{\max}, \vec{c}) = \sum_{k=1}^{k_{\max}} c_k \frac{\sum_{\substack{j=\max(0,k-N+n) \\ j \text{ odd}}}^{\min(n,k)} \binom{n}{j}\binom{N-n}{k-j}}{\binom{N}{k}} \tag{6}$$

In this equation, j represents the number of excitations included in the set S. This number can never be smaller than the size of the set minus the number of sites which are not excited. In addition, j can't be larger than the total number of excitations in the state or the size of the set. This explains the bounds on the summation over j. Also, note that only odd j values are included in the sum. This is due to the fact that only for an odd number of excitations in the set, the term $(1 - X_S)$ will contribute to the energy.

The numerator of the fraction gives the number of terms contributing to the energy[2]. The number of sets of k modes selected out of a total of N modes containing j excitations when the total number of excitations in the state is n is given by $\binom{n}{j}\binom{N-n}{k-j}$. Indeed, we select j excitations in the total of n excitations and select $k - j$ modes, since in total we need to select a set of k elements, out of the $N - n$ unexcited modes.

[2]The numerator counts the subsets S that contribute one unit of energy. Indeed, $(1-X_S)/2$ has eigenvalue one when S contains an odd number of excited modes and eigenvalue zero otherwise.

4 Finding the Gap

In order to find the gap, we must find the state with $n \neq 0$ that minimizes the energy. We can optimize over the $\vec{c}$ in order to make this value as large as possible. The gap optimization therefore looks like

$$\max_{\vec{c}}[\min_{n}(E(n, k_{\max}, \vec{c}))]. \tag{7}$$

An expression for the lower bound is found by setting $n = 1$. This results in

$$\Delta \leq \frac{k_{\max}}{N}. \tag{8}$$